

11. Container With Most Water

Solved 

Medium

 Topics

 Companies

 Hint

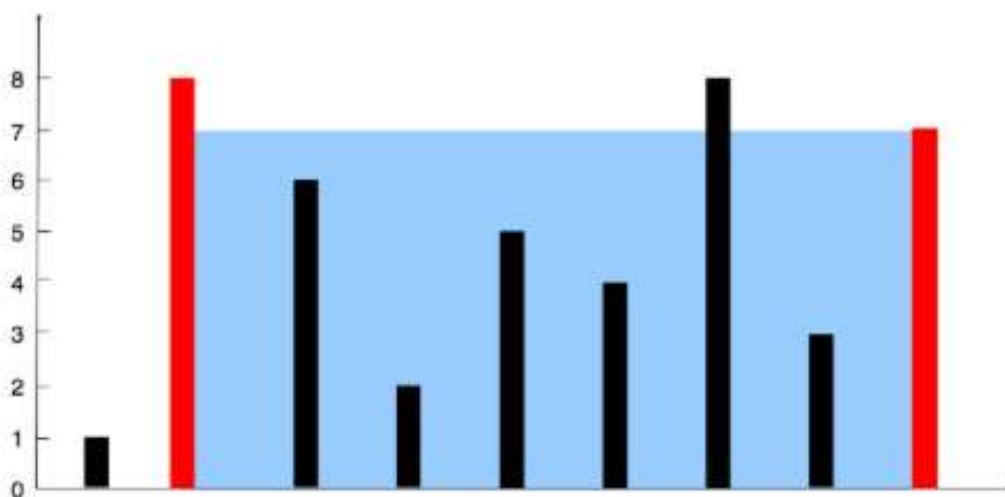
You are given an integer array `height` of length `n`. There are `n` vertical lines drawn such that the two endpoints of the i^{th} line are $(i, 0)$ and $(i, \text{height}[i])$.

Find two lines that together with the x-axis form a container, such that the container contains the most water.

Return *the maximum amount of water a container can store*.

Notice that you may not slant the container.

Example 1:



Input: `height = [1,8,6,2,5,4,8,3,7]`

Output: 49

Explanation: The above vertical lines are represented by array `[1,8,6,2,5,4,8,3,7]`. In this case, the max area of water (blue section) the container can contain is 49.

```
class Solution {
    public int maxArea(int[] height) {

        int n = height.length; int left = 0;
        int right = n-1;

        int max_area = 0;

        while(left<right){

            int h = Math.min(height[left],height[right]);

            int w = right - left;

            int area = h*w;

            max_area = Math.max(area,max_area);

            if(height[left]<height[right]){
                left++;
            }
            else{
                right--;
            }

        }

        return max_area;
    }
}
```